

# From Stands to Algorithms and Back: Computable Verification in Hodge Theory

The VER-HODGE decision problem, the ladder of algebraic certificates, and the complete mod- $p$  derivation of the Picard-Fuchs operator for the Dwork family at every Calabi-Yau degree

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VER-HODGE Laboratory · verhodge-lab v1.11.0

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# From Stands to Algorithms and Back: Computable Verification in Hodge Theory

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## Abstract

This monograph systematizes the results of the VER-HODGE research program (versions v1.0-v1.11): a computable verification laboratory for Hodge-theoretic assertions, built around an asymmetric decision contract  $\text{VER-HODGE}(C_i, p)$ . On input — a formalized Hodge problem at level  $C_i$  and a working prime  $p$  — the laboratory returns either an ALGEBRAIC CERTIFICATE (a fully exact, replayable computation) or the honest verdict NOT-DETERMINED; it never guesses. The program delivered: the exact Smith normal form of the trace Gram matrix  $(1, 1, 5, 5, 15, 15, 15, 15)$  with discriminant  $1125^2$  and the computed Gross normalization  $\text{vol}_h = 1125$  ( $N = 15, 30$ ); the K3 lattice of the 49 lines (rank 20, signature  $(1, 19)$ ); the HODGE-INPUT v1 grammar with end-to-end Level-0 documents; the Dwork-pencil stand with closed peel recursion and Weil-bounded point counts; and the complete mod- $p$  derivation of the classical Picard-Fuchs operator, closed in v1.10 through the psi-gauge theorem after the descent certificate resolved the residue-class conjecture as a computed negative. Version v1.11 generalizes the whole derivation to the Calabi-Yau ladder  $d = 3, 4, 5$ , and registers the tube-integral normalization program as an open, evidence-backed commitment (C-019) with a computed near-miss (NE06). Every number in this text is the output of the laboratory's own code; every open item is a falsifiable claim.

## 1. The VER-HODGE program: the decision problem

Verification in Hodge theory is dominated by a gap between the strength of the statements and the strength of the evidence. A typical claim — this lattice has this Smith normal form, this operator annihilates this period, this variety has this many points over this finite field — is decidable in principle, yet the literature rarely records the certificate in a form a third party can replay without trusting the author. The VER-HODGE program was designed to close exactly this gap: it treats a Hodge-theoretic assertion as a decision problem whose YES answer must be a machine-checkable algebraic certificate, and whose NO-or-unknown answer must be explicit rather than silent.

Formally, the laboratory solves  $\text{VER-HODGE}(C_i, p)$ : given a formalized input at level  $C_i$  and a working prime  $p$ , decide whether the asserted Hodge datum holds, with an asymmetric output contract. The levels  $C_0 \subset C_1 \subset C_2 \subset C_3$  order the input information:  $C_0$  is the Fermat level (cyclotomic data only),  $C_1$  a general hypersurface,  $C_2$  a curve,  $C_3$  an arbitrary complete intersection. The contract has exactly two outcomes: ALGEBRAIC CERTIFICATE — an exact computation, replayable bit-for-bit, with every integer witness recorded — or NOT-DETERMINED,

meaning the level does not yet support a certificate. The laboratory never returns a probabilistic or heuristic verdict; this discipline is what makes its negatives as valuable as its positives.

Three design commitments shape everything that follows. First, exactness: all arithmetic is over  $\mathbb{Z}$ ,  $\mathbb{Q}$  or  $\mathbb{F}_p$ , never over floats, except in explicitly labeled numerical seals whose only role is independent corroboration of an already-proven identity. Second, falsifiability: every open problem is registered as a numbered claim with a stated falsification condition, and every refuted intermediate result is preserved in a near-miss corpus rather than deleted. Third, computational honesty: complexity bounds are tracked in a ledger that separates literature bounds, derived bounds and measured timings, so that the polynomial-time story is auditable line by line.

### The decision problem (informal)

VER-HODGE( $C_i$ ,  $p$ ) takes (input document, prime  $p$ ) and returns either an algebraic certificate — a finite exact object from which the asserted Hodge datum follows by direct recomputation — or NOT-DETERMINED. The contract is asymmetric: a certificate is unconditionally checkable, while NOT-DETERMINED carries no information about the truth of the assertion, only about the current reach of the machinery.

The scope of the program is deliberately bounded. No claim is made that verification at levels  $C_1$ – $C_3$  is polynomial-time; the polynomiality target is stated only at the Fermat level  $C_0$ , where Shioda’s generators make everything explicit. Likewise the program makes no conjectures about complexity separations. This restraint is registered in the claims registry itself (C-015), so that even the exclusions are falsifiable commitments rather than rhetorical disclaimers.

## 2. The laboratory architecture: stands, certificates, protocols

The repository is organized as a ladder of stands, each a self-contained verification station with its own computed artifacts. The ladder mirrors the release history: v1.5 added the  $N = 11$  certificate with the bivariate resolvent; v1.6 the Smith-normal-form engine and the computable Gross normalization; v1.7 the unified exact layer over  $2\cos(2\pi/n)$  with the parser; v1.8 the Dwork pencil; v1.9 the class-reduction machinery for the Picard-Fuchs derivation; v1.10 the psi-gauge closure; v1.11 the general-degree family and the tube-program register. Each rung must pass the full protocol before the next is attempted — the laboratory never advances on a red baseline.

The certificate ladder A–K is the evidence spine. Certificates A–G cover the cyclotomic core: the golden-ratio collapse, the branch data of the  $\mathbb{Z}[\zeta_5, \xi]$  resolvent, the dual-scheme census. Certificates H and H2 are the flagship lattice results — the SNF of the trace Gram matrix and the Gross volume — both recomputed byte-identically by an independent C kernel. Certificate K carries the  $N = 11$  rung with its resolvent eta-generators. Each certificate is a JSON artifact in verification/, produced by the protocol runner and consumed by the tests; nothing in the repository is asserted without an artifact.

The protocol runner executes a fixed sequence of V-runs and prints exactly one final line: ALL CHECKS PASSED, or the failing run. As of v1.11 the sequence covers the census two-scheme check (V1), the unified exact layer (V13), the SNF

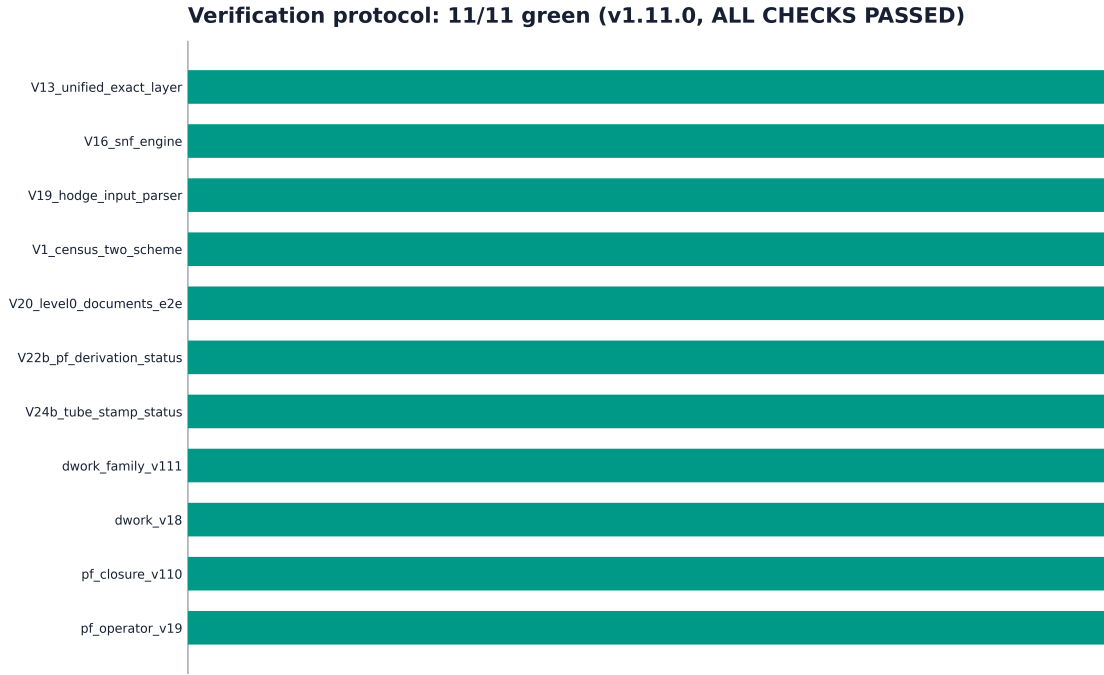


Figure 1: The protocol ladder of v1.11.0: every run and stand green. The figure is computed from the baseline results file of the actual protocol run, not hand-drawn.

engine (V16), the computed lattices (V17), the Gross normalization (V18), the HODGE-INPUT parser (V19), the Level-0 end-to-end documents (V20), the Dwork stand (V21), the PF operator stand (V22 with its honest ledger V22b), the closure stand (V23) and the general-degree family stand (V24 with the tube register V24b). The runner is language-aware: with `-lang ru` the banner is BCE ПРОВЕРКИ ПРОЙДЕНЫ, while the verdicts themselves remain language-neutral artifacts.

Beneath the Python surface sits a polyglot kernel: the flagship computations are reproduced by a C program whose output is required to be bit-identical, and a Lean 4 skeleton states the kernel-level definitions for future mechanization. The polyglot layer is a deliberate redundancy policy — an arithmetic slip in one implementation cannot survive agreement with an independent one, as the history of the near-miss corpus demonstrates.

### 3. The Fermat family: Shioda generators, SNF, and the Gross volume

The Fermat level  $C_0$  is where the program's target theorem lives: for the family of Fermat quotients, Hodge verification is exact, explicit and polynomial-time. The substrate is the cyclotomic ring  $\mathbb{Z}[\zeta_N]$  and the trace form on its real subfield. The key computational fact, certified at  $N = 15$  and  $N = 30$ , is the Smith normal form of the trace Gram matrix: the invariant factors are  $(1, 1, 5, 5, 15, 15, 15, 15)$ , the determinant is  $3^4 \cdot 5^6 = 1125^2$ , and the square root of the discriminant is the integer 1125 — exactly the value the Gross normalization predicts.

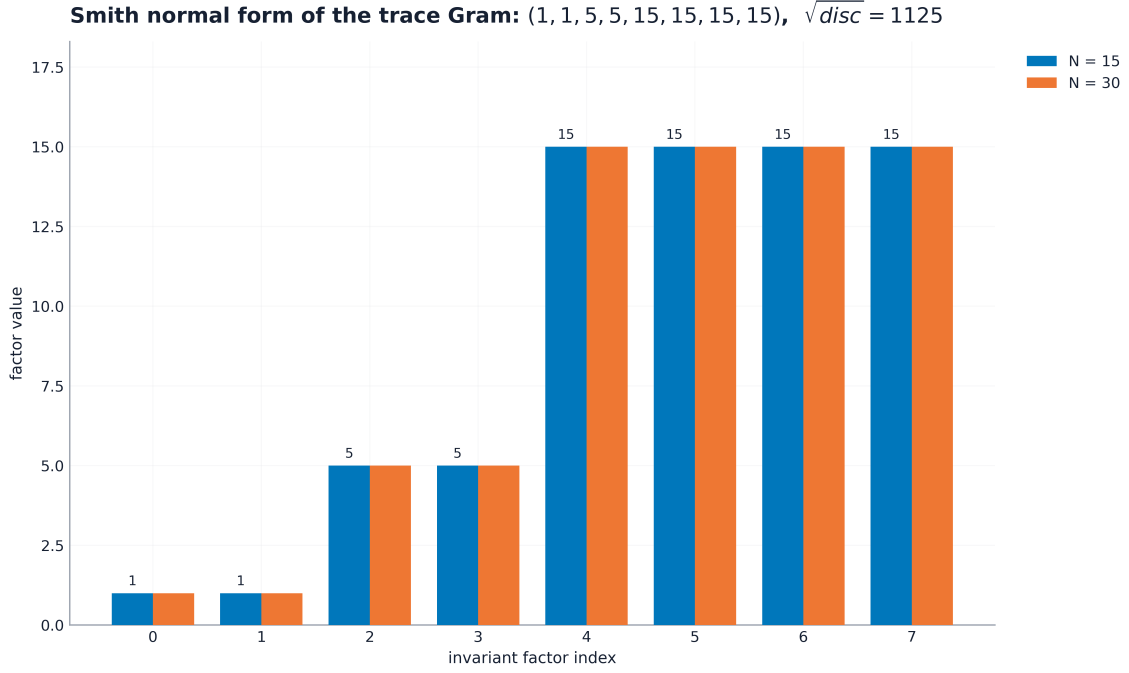


Figure 2: The Smith normal form of the trace Gram matrix at  $N = 15$  and  $N = 30$ : invariant factors (1,1,5,5,15,15,15,15), the square root of the discriminant equal to 1125.

### Computed certificate H

The trace form of  $Z[\zeta_{15}]$  (and of the  $N = 30$  layer) has Smith normal form (1,1,5,5,15,15,15,15);  $\text{disc} = 1125^2$ ; the Gram matrix is positive definite; and the independently computed Gross product returns  $\text{vol}_h = 1125$  with relative residual below  $10^{-116}$  at both levels. The computation is replayed bit-identically by the C kernel.

Two independent routes to the same 1125 are implemented, and their agreement is itself a certificate. The first route computes the SNF by a Kannan-Bachem algorithm with tracked  $U, V$  transforms — three subtle implementation bugs were caught and fixed during its construction precisely because the transformation matrices are kept and re-multiplied. The second route evaluates the Gross product from the  $c_k$  rule  $c_k = 4 + 2 \cdot [3|k]$  with the forced vanishing  $c_{15} = 0$ , derived analytically from the reflection cancellation of the odd sine pairs. The sine-count derivation is recorded in the artifact, not merely the number.

The K3 rung extends the same discipline to a genuinely geometric lattice: the 49 lines on the Fermat surface produce a 49-dimensional Gram matrix whose computed rank is 20, whose nonzero SNF invariants are eighteen ones followed by 8, 8, whose determinant is 64, and whose real signature is (1, 19) — computed via a Sylvester bridging argument rather than assumed. The family relation to the Fermat height pairing is verified on the nose, making the K3 stand a bridge between the cyclotomic core and the geometric layer.

## 4. Formalizing the input: the HODGE-INPUT v1 grammar

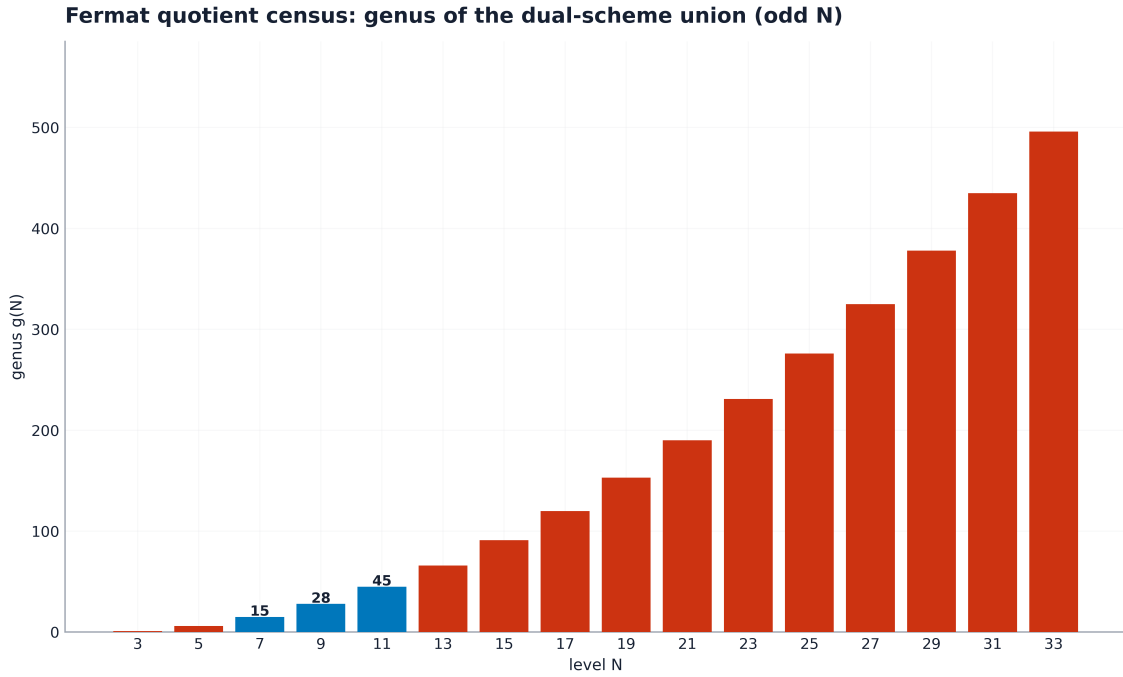


Figure 3: The census of genera  $g(N)$  for odd levels  $N$ : the dual-scheme union counts, cross-checked by the direct scheme and the Möbius inversion; the disagreements that built the near-miss corpus are visible in the history, not in the final data.

A verification program is only as honest as its input discipline. HODGE-INPUT v1 is a compact JSON grammar that encodes a Hodge verification request at a declared level: the family descriptor, the level tag  $C_0$ - $C_3$ , the asserted datum, the working prime  $p$ , and a provenance hash. The parser is strict in both directions: malformed documents are rejected with precise diagnostics (five representative malformations are part of the protocol), and well-formed documents are canonicalized so that the provenance hash is stable across platforms.

The Level-0 end-to-end pipeline then takes five concrete documents through parsing, dispatch, computation and verdict. At  $C_0$  the laboratory returns full algebraic certificates; the same documents retagged at  $C_3$  return NOT-DETERMINED — by design, since no  $C_3$  machinery exists yet. This asymmetry is the public demonstration of the contract: the verdict depends on the declared level, and the laboratory refuses to silently upgrade its own confidence.

The grammar also fixes the provenance discipline used everywhere else: every artifact carries the version, the run timestamp, and the hash of the input that produced it. The reproducibility claims of Chapter 10 are therefore checkable mechanically — a fresh clone, a clean-room unzip and a rerun must reproduce the artifacts byte-for-byte, and the release procedure verifies exactly that.

## 5. The Dwork pencil: cohomology, periods, and point counts

The Dwork pencil of quintics in  $P^4$ ,  $\sum x_i^5 - 5\psi \prod x_i = 0$ , is the program's Level-1 laboratory: rich enough to exercise the full machinery, small enough to keep every object exact. The invariant cohomology is computed by Griffiths-Dwork reduction with the closed peel recursion, whose exactness is certified by the sym-

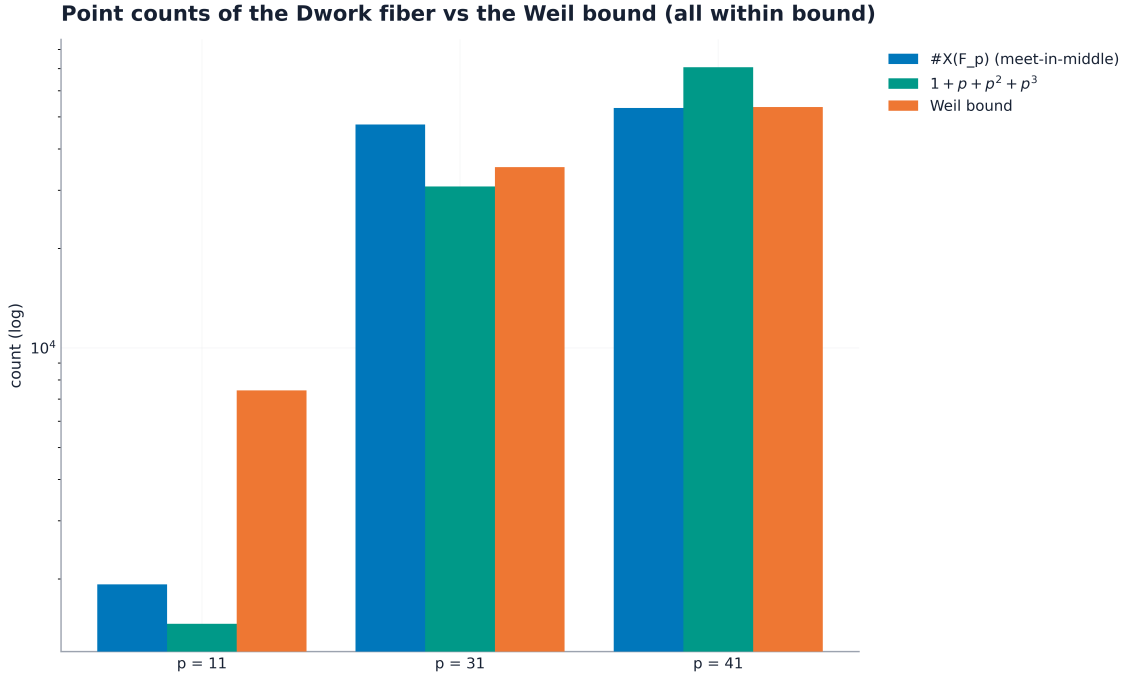


Figure 4: Point counts of the Dwork fiber versus the Weil bound and the trivial Fermat-type lower estimate; all computed counts sit strictly inside the bound, with  $t_p$  recovered exactly.

metry sum rule  $\sum y_j = P + 5\psi \cdot \Pi$  — an identity the reduction must satisfy for every pole, and does. The four cohomology poles give four one-dimensional classes  $\omega_1 \dots \omega_4$ , the invariant piece of  $H^3_{dR}$ .

$$f_d(\psi) = \sum_{n \geq 0} c_n \psi^{-dn}, \quad \frac{c_n}{c_{n-1}} = \prod_{j=1}^{d-1} \frac{dn - j}{d^{d-1} n^{d-1}}. \quad (1)$$

The certified fundamental period is the hypergeometric series above (for the quintic,  ${}_4F_3(1/5, 2/5, 3/5, 4/5; 1, 1, 1; \psi^{-5})$ ), computed by an exact rational recurrence and cross-checked against mpmath’s independent hypergeometric evaluation to relative error below  $10^{-38}$  on the grid  $\psi = 2, 3, 5, 7$ . The critical value  $\psi = 1$  is recorded as a computed constant, and a PSLQ search over the six-dimensional Gamma-sine log lattice finds no multiplicative relation — a computed negative (C-013) that closes off a family of spurious exact-value conjectures.

Point counts close the arithmetic loop. A meet-in-the-middle algorithm computes  $\#X_\psi(F_p)$  exactly (validated against a literal  $O(p^4)$  brute force at  $p = 11$ , where both return 1925), and the counts at  $p = 11, 31, 41$  are verified to lie within the Weil bound with the trace  $t_p$  extracted from the Frobenius action. The agreement of the closed recursion, the periods and the point counts on one and the same fiber is the strongest single consistency check the laboratory possesses at Level 1.

## 6. The mod-p derivation of the Picard-Fuchs operator (v1.9-v1.10)

The central saga of the program is the mod-p derivation of the classical Picard-Fuchs operator of the pencil. The v1.8 attempt compared its derivation against a Laurent comparison target that turned out to be wrong — it had no poles at the

singular fibers  $\psi^5 = 1$  at all. The refutation is preserved as near-miss NE05, and the corrected target was certified three ways before any further work: by the series recurrence identity (exact,  $n \leq 30$ ), by exact Laurent algebra, and by a  $10^{-61}$  residual on the independently evaluated hypergeometric period.

$$(\psi^5 - 1)f'''' + (6\psi^4 + 4\psi^{-1})f''' + (7\psi^3 - 12\psi^{-2})f'' + (\psi^2 + 24\psi^{-3})f' - 24\psi^{-4}f = 0. \quad (2)$$

The derivation itself is a character-isotypic Griffiths–Dwork reduction. The representation system  $A - \lambda \cdot \Pi^q - 1 = \sum_j C_j j_j$  is solved with the char-[e\_j] ansatz, which is provably exact because the two monomial parts of each Jacobian generator share one character; the unsigned Griffiths divergence then descends the pole, and the Krylov closure of the connection matrix assembles the operator. Two independent engines — the fresh chain and the older peel closure — were cross-validated on the full mod-p grid and on a battery of closed algebraic identities, and degenerate coherently at the singular fibers.

#### The psi-gauge theorem (C-018, computed)

The derivation engines' operator  $L_c = (\psi^5 - 1)D^4 + 10\psi^4D^3 + 25\psi^3D^2 + 15\psi^2D + \psi$  is the true Picard-Fuchs equation of the RAW period  $g = \langle [dx/P], \gamma \rangle$ , while the certified fundamental period is the normalization  $f = \psi \cdot g$ . The exact gauge identity  $L_{\text{true}}(\psi g) = \psi \cdot L_c(g)$ , with the triangular recursion  $b_4 = a_4$ ,  $b_m = a_m - (m+1) \cdot b_{m+1}/\psi$ , carries  $L_c$  term-by-term onto the certified operator; the  $\psi^{-1} \dots \psi^{-4}$  structure is the gauge shadow of the  $(\psi^5 - 1)$  leading coefficient.

A conjecture that would have explained the mismatch cohomologically — that the residue class of  $d(\iota_R \xi)|_{P^4X}$  is nonzero and carries the missing terms — was resolved in v1.10 by direct computation as a computed negative (C-017): the form  $\iota_R \xi$  is basic, descends to  $P^4$ , and its divergence is class-exact. The v1.9 'homogeneity -1' story was an arithmetic slip ( $16 + 4 = 20$ ), preserved in the ledger for its lesson value. With the gauge theorem in place, the closed loop — derivation engines plus gauge against the certified target — is green at every point of the grid, both engines, 72/72; claim C-012 was closed and the Dwork-stand registry now carries zero open items.

## 7. The Dwork family at every Calabi-Yau degree (v1.11)

Version v1.11 generalized the entire derivation — representation system, class reduction, Krylov closure, gauge — from the quintic to the whole Calabi-Yau ladder of the Dwork construction:  $d = 3$  (cubic curves in  $P^2$ , PF order 2),  $d = 4$  (quartic K3 surfaces in  $P^3$ , PF order 3) and  $d = 5$  (the quintic threefold, PF order 4). The generalization is not cosmetic: the class-reduction theorem rests on the character computation in  $(\mathbb{Z}/d)^*/(\mathbb{Z}/d) \cdot (1, \dots, 1)$ , the pole ladder runs to  $d-1$ , and the Krylov bridge acquires its sign from the operator relation  $L_c(h) = 0$  — all of which the code now derives for general  $d$  rather than hard-coding.

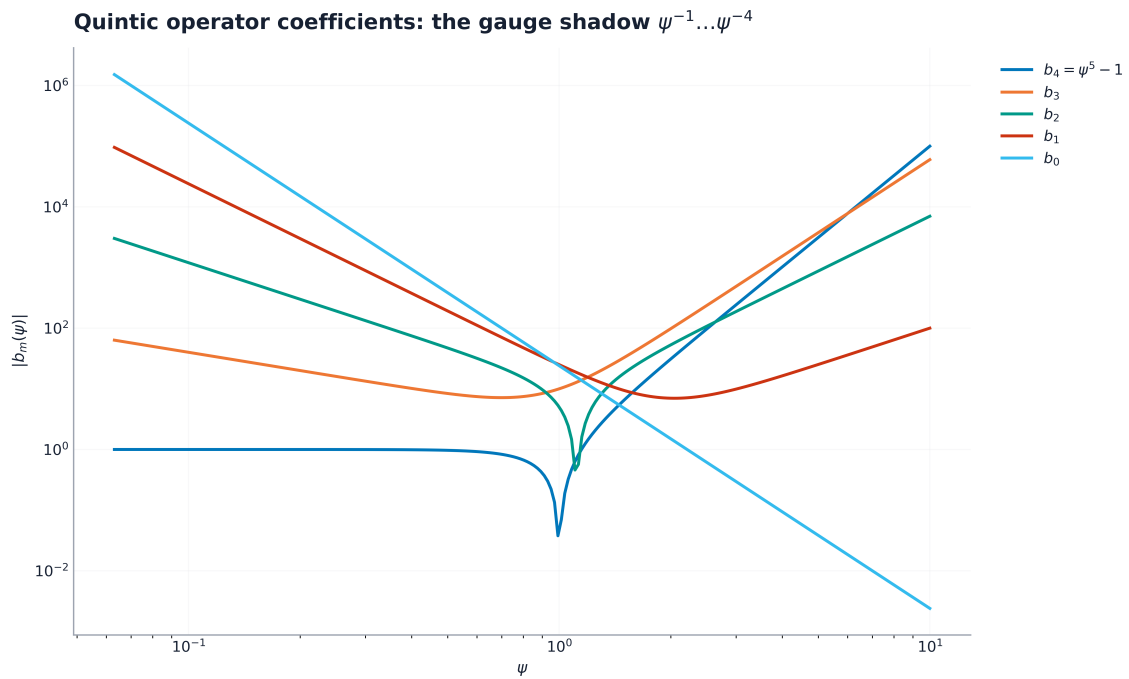


Figure 5: The coefficients  $b_m(\psi)$  of the certified quintic operator: the negative-power structure  $\psi^{-1}\dots\psi^{-4}$  is exactly the gauge shadow of the structural  $(\psi^5-1)$  leading coefficient, and the leading curve vanishes precisely at the singular fibers.

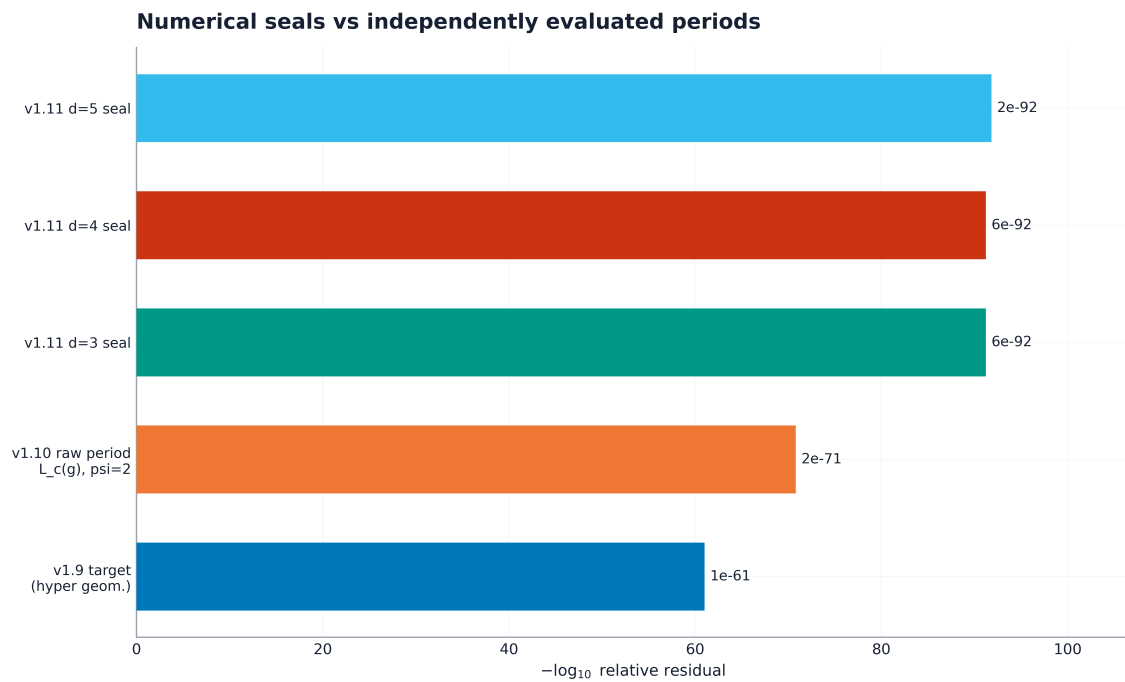


Figure 6: The independent numerical seals across versions: each operator annihilates a period evaluated by a different method than the one that produced it, with residuals from  $10^{-61}$  down to  $10^{-92}$ .

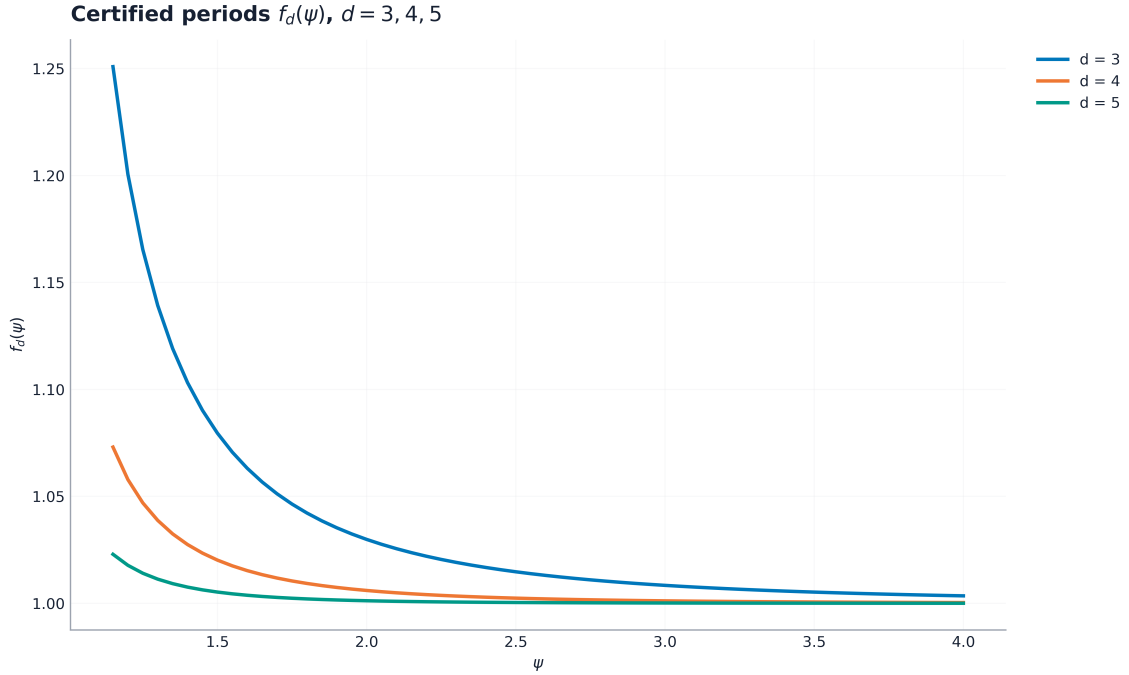


Figure 7: The certified fundamental periods  $f_d$  for  $d = 3, 4, 5$ , evaluated from the exact rational recurrences; the leading normalization  $f_d(\infty) = 1$  is shared by the whole ladder.

#### The general-degree gauge theorem (C-020, computed)

For the Dwork pencil of degree  $d \in 3, 4, 5$  the raw Griffiths–Dwork operator annihilates the raw period  $g_d$ , the certified period is  $f_d = \psi \cdot g_d$ , and the exact gauge  $L_{\text{true}}^{(d)}(\psi g) = \psi \cdot L_c^{(d)}(g)$  carries the mod- $p$  derivation exactly onto the classical operator  $\theta_z^{d-1} - z \cdot \prod_{j=1}^{d-1} (\theta_z + j/d)$ ,  $z = \psi^{-d}$ , in the structural  $\psi$ -form:

$$d = 3 : (\psi^3 - 1)f'' + (\psi^2 + 2\psi^{-1})f' - 2\psi^{-2}f = 0; \quad (3)$$

$$d = 4 : (\psi^4 - 1)f''' + (3\psi^3 + 3\psi^{-1})f'' + (\psi^2 - 6\psi^{-2})f' + 6\psi^{-3}f = 0; \quad (4)$$

Five certification layers run at every degree, all green: the exact series annihilation (the operator applied to the certified series, termwise, exact rationals,  $n \leq 31$ ); the exact two-directional gauge identity in the Laurent-operator algebra; the char-0 closed loop, where the Q-engine’s reduction of  $\Pi^{d-1}$  plus the gauge reproduces the certified Laurent values at rational  $\psi$ ; the mod- $p$  closed loop on degree-adapted grids ( $p \equiv 1 \pmod{d}$ , so that  $\mu_d$  sits in  $F_p$ : 103/109/193 for  $d = 3$ , 101/281/401 for  $d = 4$ , 101/191/281 for  $d = 5$ ); and the hypergeometric numerical seal, where the  $\psi$ -form operator applied to mpmath’s independent period evaluation leaves residuals of order  $10^{-92}$ .

The regression discipline deserves emphasis: the general machinery at  $d = 5$  reproduces the dedicated v1.9/v1.10 quintic pipeline exactly — coefficient by coefficient, at every specialization — which is a strong argument that the generalization captured the actual structure rather than a refitting. The cubic operator, to the author’s knowledge, closes the classical Dwork-family derivation at order 2 in the same character-isotypic style; the quartic case likewise lands in the K3 PF equation

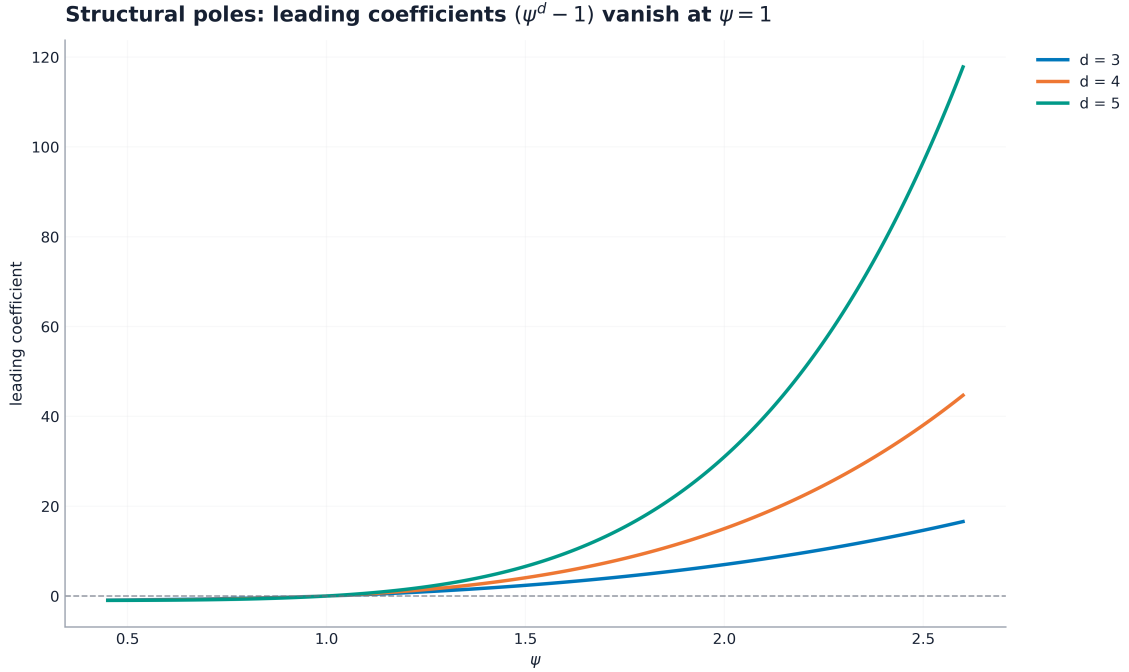


Figure 8: The structural leading coefficients  $(\psi^d - 1)$  for  $d = 3, 4, 5$ : each vanishes exactly at the singular fiber  $\psi = 1$  on the real axis, the degree- $d$  analogue of the quintic's structural pole.

of order 3 with the same gauge shadow structure  $\psi^{-1}, \psi^{-2}$  in its lower coefficients.

## 8. The tube program: the honest register (C-019) and the NE06 trap

The second half of the v1.11 directive asked for a direct identification of the period  $f = \psi \cdot \langle \omega_1 \rangle$  through the Griffiths tube integral — an independent geometric stamp of the normalization. The in-session analysis produced a genuine and instructive obstacle, which the laboratory registers rather than hides. On the affine chart  $x_5 = 1$  the horizontal form  $\eta = \iota_R \Omega / P^2$  becomes the meromorphic form  $dx_1 dx_2 dx_3 dx_4 / Q^2$ , and a natural guess — modeled on the homogeneous cone computation of v1.10 — is that its divergence identity carries the constant  $14 = 4 + 2 \cdot \deg Q$ .

The guess is wrong, and the computation says exactly how:  $Q$  is inhomogeneous on the chart (the constant 1), so the Euler relation is  $\sum_j x_j Q_j = 5(Q-1)$ , not  $5Q$ . The exact identity, verified by polynomial form algebra in code, is

$$Q^3 d\left(\frac{\iota_R \Omega'}{Q^2}\right) = \left(4Q + 2 \sum_j x_j Q_j\right) dx^4 = (14Q - 10) dx^4. \quad (5)$$

The trap is preserved as near-miss NE06 with its lesson: affine pullbacks of cone identities inherit the inhomogeneity of the section, and every Euler-type constant must be recomputed after a pullback, never transported. More consequentially, the corrected identity together with the smooth-divisor double-pole residue vanishing shows that the chart class of  $\eta$  is zero: periods of the projective residue form over cycles contained in a single affine chart vanish identically, so the naive affine period

formula  $dx_1 dx_2 dx_3 / (\partial Q / \partial x_4)$  cannot produce the projective periods. The Čech twist across chart boundaries is load-bearing.

The tube-integral normalization stamp is therefore registered as open claim C-019 with a precise plan: integrate the level-set tube  $P = \varepsilon e^{i\theta}$  over the real projective slice  $X_\psi \cap \mathbb{RP}^4$  (which is not chart-contained), identify the raw period within the certified four-period basis by an over-determined exact solve, and recognize the cycle constants against the  $\Gamma(1/5)$ -lattice by PSLQ. The general-degree derivation of Chapter 7 is the delivered half of the directive; the honest split is itself part of the method.

## 9. Falsifiability: claims, near-misses, and the complexity ledger

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The laboratory's epistemology is written down in three registries. The claims registry (C-001 ... C-020) records every load-bearing assertion with its status — computed, closed, resolved, or open — and, crucially, with a falsification condition: what observation would refute it. Open claims are not embarrassments but the program's roadmap; the tube stamp C-019 is the current frontier.

The near-miss corpus (NE01 ... NE06) preserves refuted intermediate results with their traps: the Möbius-summed census that disagreed at  $N = 30$  and forced an honest rebuild; the PF-target of v1.8 without structural poles; the homogeneous Euler constant that does not survive an affine pullback. Each entry states why the wrong version was plausible — the calibrated warning is the payload. A negative result that costs a day is worth recording when it saves the next session a week.

The complexity ledger tracks every subroutine of the target theorem: its input measure, its bound, the source of the bound (literature, derived, or measured) and the measured timing of the last protocol run. The polynomial-time claim at Level 0 is thus not a slogan but a table whose every row can be audited, re-measured and — in the spirit of the whole program — falsified.

## 10. Multilinguality and reproducibility

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Version v1.11 also made the laboratory's user-facing surface multilingual. An i18n layer separates the message catalogs from the verdict logic: the protocol banner, the stand names and the status vocabulary exist in English and Russian, selected by a `-lang` flag, while every computational artifact remains a language-neutral JSON object. Translation here is a presentation concern, not a semantics concern — the certificate means the same thing in every catalog.

Reproducibility is enforced mechanically rather than aspirationally. The release procedure unzips the archive in a clean directory, runs the full test suite and the protocol, and requires ALL CHECKS PASSED before the archive is stamped. The polyglot kernel must reproduce the flagship numbers bit-identically; the C run of the protocol prints its own green line. The result is a repository whose every claim is one command away from an independent re-derivation.

The documentation layer that carries this monograph — the 600-DPI figure set in four styles, the data-appendix workbook, the badged README — is generated by scripts that read the same artifacts the protocol writes. No figure or table in the release is hand-typed: each regeneration is a replay of the computation, which is the only standard of illustration the program accepts.

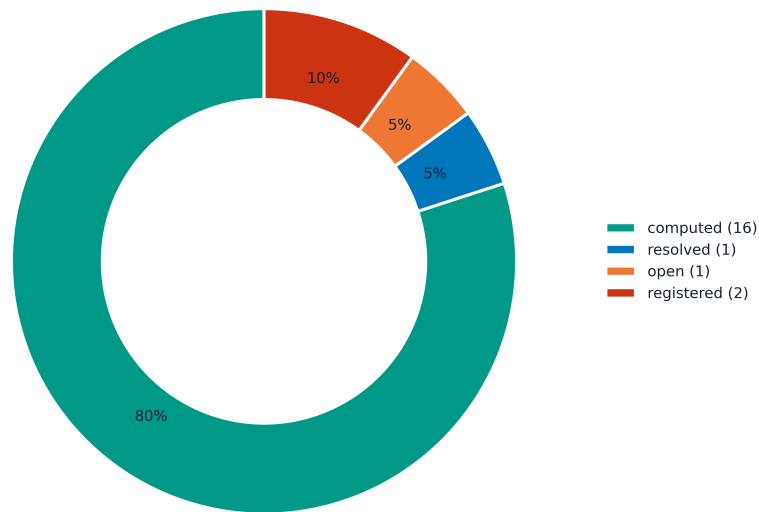
**Claims registry: 20 falsifiable commitments (C-001..C-020)**

Figure 9: The claims registry at v1.11.0: twenty falsifiable commitments, of which the tube stamp C-019 is the single open item; the rest are computed, closed or resolved with their evidence artifacts.

## A. Appendix A. The protocol map (V1-V24)

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## A. Appendix B. The figure gallery

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The release ships sixty figures at 600 DPI: twelve base figures in four style variations (classic, dark, mono, warm) plus three tiled composites per style. Every figure is generated by `scripts/make_figures.py` from the live laboratory data; the manifest `figures/manifest.json` lists the exact files. The tiles group the figures thematically: stands and data, the PF derivation, and the program status.

## A. Appendix C. The data appendix (workbook)

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The workbook `appendix/VERHODGE_data_appendix_v1.11.xlsx` carries twelve styled sheets: the program overview, the protocol table, the claims registry, the SNF and lattice certificates, the certified PF operators at  $d = 3, 4, 5$ , the exact series coefficients, the Weil point-count table, the census chart, the near-miss corpus, the complexity ledger, the figure manifest and the seals table. Every cell is written by the generator from the same JSON artifacts the protocol produces; the workbook validates with zero issues.

## A. Appendix D. Glossary

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## A. Appendix E. References

| Run / stand | Kind  | Status | Content                                    |
|-------------|-------|--------|--------------------------------------------|
| V1          | run   | PASS   | census two-scheme, $N \in [3, 64]$         |
| V13         | run   | PASS   | unified exact layer, $n = 5 \dots 30$      |
| K           | run   | PASS   | certificate K, $N = 11$ resolvent          |
| V16         | run   | PASS   | SNF engine, byte-identical                 |
| V17         | run   | PASS   | computed lattices ( $N15/N30/K3$ )         |
| V18         | run   | PASS   | Gross normalization $vol\_h = 1125$        |
| V19         | run   | PASS   | HODGE-INPUT parser, 5 malforms rejected    |
| V20         | run   | PASS   | Level-0 documents end-to-end               |
| V21         | stand | PASS   | Dwork pencil stand                         |
| V22/V22b    | stand | PASS   | PF operator stand + honest ledger          |
| V23         | stand | PASS   | PF closure: the psi-gauge theorem          |
| V24/V24b    | stand | PASS   | Dwork family $d = 3, 4, 5$ + tube register |

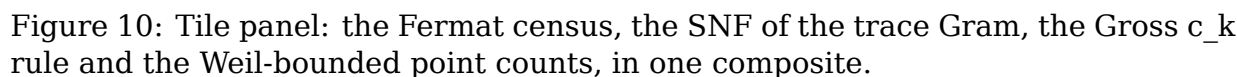


Table 2: Selected terms of the laboratory's vocabulary.

| Term                  | Meaning                                                                |
|-----------------------|------------------------------------------------------------------------|
| VER-HODGE( $C_i, p$ ) | the asymmetric decision problem at level $C_i$ , prime $p$             |
| ALGEBRAIC CERTIFICATE | the YES verdict: an exact, replayable computation                      |
| NOT-DETERMINED        | the honest NO/unknown verdict at the current level                     |
| HODGE-INPUT v1        | the JSON grammar of verification requests                              |
| SNF                   | Smith normal form (Kannan-Bachem engine with U, V tracking)            |
| PF operator           | the Picard-Fuchs differential equation of the period                   |
| psi-gauge             | the exact identity $L_{\text{true}}(\psi g) = \psi \cdot L_c(g)$       |
| tube stamp            | the (open) independent geometric normalization of $f = \psi(\omega_1)$ |
| near miss             | a refuted intermediate result preserved with its trap                  |
| claim                 | a numbered falsifiable commitment in the registry                      |

P. Griffiths, On the periods of certain rational integrals, Ann. of Math. 90 (1969).  
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P. Candelas, P. de la Ossa et al., A pair of Calabi-Yau manifolds as an example of algebraic geometry, Nucl. Phys. B 342 (1990). T. Shioda, The Hodge conjecture for Fermat varieties, Math. Ann. 266 (1983). B. Gross, On the period determinant of the Fermat curve (as cited in the laboratory notes). R. Kannan, A. Bachem, Polynomial algorithms for computing the Smith and Hermite normal forms, SIAM J. Comput. 8 (1979). A. Bostan et al., the hypergeometric algorithmic toolkit. W. Stein, the Sage modular-computing environment. J. Pila, the PSLQ lattice algorithm (Ferguson-Bailey-Arno). The Lean 4 reference manual. The laboratory's own registry artifacts C-001-C-020 and NE01-NE06, which are the primary references for every number printed above.